



**Verified Carbon  
Standard**

## BLITTA – TOGO SOLAR POWER PLANT

Document Prepared by ELEMENT 6 – FZCO

ON BEHALF OF CLIMATEPARTNER GmbH



|                          |                                                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Title</b>     | BLITTA – TOGO SOLAR POWER PLANT                                                                                                                                                                              |
| <b>Version</b>           | 6                                                                                                                                                                                                            |
| <b>Report ID</b>         | 3298 - I                                                                                                                                                                                                     |
| <b>Date of Issue</b>     | 15-August-2023                                                                                                                                                                                               |
| <b>Project ID</b>        | 3298                                                                                                                                                                                                         |
| <b>Monitoring Period</b> | 12-June-2021 to 31-December-2022 (including both dates)                                                                                                                                                      |
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# 1 PROJECT DETAILS

## 1.1 Summary Description of the Implementation Status of the Project

The Blitta - Togo solar PV plant is one of the largest in West Africa and the first renewable energy facility in the country<sup>1,2</sup>. This plant, which is located in Blitta, supplies reliable, clean electricity to the national grid which then supplies to hundreds of thousands of homes and businesses in the country and save more than one million tonnes of CO<sub>2</sub> emissions through its lifetime. The current capacity of the plant is 50 MW. An additional 20 MW expansion is expected to be operational by 2024 bringing the total to 70 MW. The commissioning date of Phase 1 was on 12-June-2021 while the commissioning date of Phase 2 is planned for 01-December-2023.

The village of Blitta-Losso is part of the canton of Blitta village and has an average monthly direct normal radiation of 5.14 kWh/m<sup>2</sup>/day and an average annual solar irradiation of 1,876 kWh/m<sup>2</sup>. The coordinates of the project location are 8°21'52.33"N, 1°1'4.42"E. The project will advance the Togolese national clean energy strategy to increase the share of renewables in electrification to 50 per cent by 2025 and 100 per cent by 2030. The current monitoring period is from 12-June-2021 to 31-December-2022 (including both dates). The project generated 123,855.67 MWh of electricity for the current monitoring period. The total GHG emission reductions or removals generated in this monitoring period are 70,969 tCO<sub>2</sub>e. The project activity, which has a lifetime of 25 years, produced an average of 6,518.72 MWh/month of electricity for the given monitoring period.

| Audit Type   | Period                                                  | Program | WVB Name                                  | Number of years |
|--------------|---------------------------------------------------------|---------|-------------------------------------------|-----------------|
| Validation   | 12-April-2023                                           | VCS     | LGAI TECHNOLOGICAL CENTER, S.A. (APPLUS+) | N/A             |
|              | Year 1                                                  |         |                                           |                 |
| Verification | 12-June-2021 to 31-December-2022 (including both dates) | VCS     | LGAI TECHNOLOGICAL CENTER, S.A. (APPLUS+) | 1.5             |
|              | Year 1                                                  |         |                                           |                 |
| Total        |                                                         |         |                                           | 1.5             |

<sup>1</sup> <https://www.irena.org/newsroom/pressreleases/2021/Jun/Togo-Inaugurates-50MW-Solar-Plant-Financed-Under-IRENA-ADFD-Facility>

<sup>2</sup> <https://www.mofaic.gov.ae/en/mediahub/news/2021/6/24/24-06-2021-uae-togo>

## 1.2 Sectoral Scope and Project Type

Sectoral Scope: 1, Energy industries (renewable -/non-renewable sources)

Type: Renewable energy projects (Large Scale)

Category: ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0

The following tools referred by the methodology have been used and their applicability is discussed in section 3.2.

- TOOL01: Tool for the demonstration and assessment of additionality --- Version 7.0.0
- TOOL07: Tool to calculate the emission factor for an electricity system --- Version 7.0

The project is a renewable energy type and is not a grouped project.

## 1.3 Project Proponent

|                          |                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Organization name</b> | AMEA Solar Togo SAU                                                                                            |
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| <b>Title</b>             | Senior Director – Project Development                                                                          |
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|                          |                                                  |
|--------------------------|--------------------------------------------------|
| <b>Organization name</b> | ClimatePartner GmbH                              |
| <b>Contact person</b>    | Dr. Sascha Lafeld                                |
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| <b>Email</b>             | sascha.lafeld@climatepartner.com                 |

## 1.4 Other Entities Involved in the Project

|                     |                                                    |
|---------------------|----------------------------------------------------|
| Organization name   | Element 6 – FZCO                                   |
| Role in the Project | Consultant                                         |
| Contact person      | Amer Arafat                                        |
| Title               | Partner                                            |
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| Telephone           | +971 52 938 9590                                   |
| Email               | amer@element6.cc                                   |

## 1.5 Project Start Date

The project start date was on 12-June-2021.

## 1.6 Project Crediting Period

The project's crediting period is 7 years renewable.

The first crediting period starts from 12-June-2021 and lasts till 11-June-2028.

## 1.7 Project Location

The plant is located in Blitta, Togo. The village of Blitta-Losso is part of the canton of Blitta village and is surrounded from north to south and from east to west by the villages of Blitta kotokoli, Boufouli Boko Losso, Doufouli and Oranyi. The coordinates of the project location are 8°21'52.33"N, 1°1'4.42"E.



Figure 1 - Site location for the 70MW PV Power Plant

## 1.8 Title and Reference of Methodology

The methodology applied is:

- ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0

The following tools referred by the methodology have been used:

- TOOL01: Tool for the demonstration and assessment of additionality --- Version 7.0.0
- TOOL07: Tool to calculate the emission factor for an electricity system --- Version 7.0

## 1.9 Participation under other GHG Programs

The project is neither registered nor seeking registration under any other GHG programs.

## 1.10 Other Forms of Credit and Supply Chain (Scope 3) Emissions

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. The project has not sought or received another form of GHG-related credit, including renewable energy certificates. The project does not affect Scope 3 emissions associated with any good or service.

## 1.11 Sustainable Development Contributions

AMEA Solar Togo SAU has put in place the following CSR activities across essential sectors:

- Construction of a clinic in the village of Blitta for curative and preventive service

- Distributed essential food and hygiene products to 2,200 families as a response to the lockdown in Togo
- Construction and renovation of primary schools
- Provision of necessary furniture and educational equipment
- Internship program for 36 students from Lomé Technical University
- Construction and commissioning of the Blitta Solar Plant
- Temporary jobs created during the construction of the plant and permanent jobs created for the operation and maintenance of the plant
- Reduction of total CO<sub>2</sub> emissions generated.

The evidence of the project's SD contributions is provided in Appendix I.



Table 1: Sustainable Development Contributions

| Row number | SDG Target | SDG Indicator                                                     | Net Impact on SDG Indicator        | Current Project Contributions                                                                                                                                                                                                                                                                                                                                                                                                       | Contributions Over Project Lifetime                                                                                                                                      |
|------------|------------|-------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1)         | 3.8        | Number of patients visiting the newly constructed clinic per year | Implemented activities to increase | <p>AMEA Solar Togo SAU built the first clinic in Blitta, and this has improved the standard of living within the village.</p> <p>The number of patients visiting the newly constructed clinic increased as there was no clinic previously.</p> <p>Since the opening of the clinic, 1,576 patients have visited the clinic. (1,321 medical consultations, 249 Prenatal consultations.) There has also been a total of 45 births.</p> | Throughout the project lifetime, approximately 30,000 patients are expected to visit the clinic. Please refer to Appendix I for details on patient visits to the clinic. |

|    |     |                                                      |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                 |
|----|-----|------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2) | 4.1 | Number of students going to primary schools annually | Implemented activities to increase | <p>AMEA Solar Togo SAU has built three (3) primary schools and renovated the four (4) existing primary schools with the necessary equipment and furniture.</p> <p>The number of students going to primary schools annually has increased.</p> <p>EPP BLITTA CENTRAL D (TOMEGBE):</p> <ul style="list-style-type: none"> <li>• 2020-2021: 145 students</li> <li>• 2021-2022: 131 students</li> <li>• 2022-2023: 158 students</li> </ul> <p>LYCEE BLITTA CARREFOUR:</p> <ul style="list-style-type: none"> <li>• 2021-2022: 200 students</li> <li>• 2022-2023: 379 students</li> </ul> <p>EPP KOKONADE</p> <ul style="list-style-type: none"> <li>• 2021-2022: 102 students</li> <li>• 2022-2023: 94 students</li> </ul> | The number of students going to primary schools is estimated to be around 30,000 over the entire project lifetime. Please refer to Appendix I for details on student admissions to the schools. |
|----|-----|------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|    |     |                                                                             |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                  |
|----|-----|-----------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |     |                                                                             |                                    | <p>EPP KATAKPE</p> <ul style="list-style-type: none"> <li>• 2021-2022: 87 students</li> <li>• 2022-2023: 94 students</li> </ul> <p>EPP LAMAKAZA B</p> <ul style="list-style-type: none"> <li>• 2020-2021: 136 students</li> <li>• 2021-2022: 180 students</li> <li>• 2022-2023: 203 students</li> </ul>                                                                                                                                    |                                                                                                                                                                                  |
| 3) | 7.b | Total clean energy generated from the Blitta Solar Plant. Measured in (MWh) | Implemented activities to increase | <p>The project has provided electricity and energy to the village of Blitta which had no access to electricity prior to the plant. The project also feeds energy into the National Grid. It is also the first renewable energy plant in the country.</p> <p>The total clean energy produced in the country has increased since the completion of this project. The project generated 123,855.67 MWh for the current monitoring period.</p> | The project is expected to generate 2,921 GWh for the entire project lifetime. The invoices for energy generation for the current monitoring period are attached to this report. |

|    |      |                                                                                      |                                    |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                 |
|----|------|--------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4) | 8.1  | Number of employees working in the operation and maintenance activities of the plant | Implemented activities to increase | There are currently 13 employees that work in the operations and maintenance of the plant. Seven (7) technicians, one (1) plant manager, one (1) SCADA engineer, one (1) field engineer, one (1) HSE employee, one (1) driver and one (1) housekeeper. | <p>The project created employment for 13 employees that work in the operations and maintenance of the plant. Please refer to the Organization chart in Section 4.3.</p> <p>Based on the above, there will always be 13 people employed in the plant for the entire lifetime of the project.</p> |
| 5) | 13.0 | Tonnes of greenhouse gas emissions avoided or removed (tCO <sub>2</sub> e/year)      | Implemented activities to increase | The renewable energy plant has prevented the release of 70,969 tCO <sub>2</sub> e of greenhouse gas emissions for the current monitoring period.                                                                                                       | The project is expected to prevent the release of 1,674 kilotonnes of carbon into the atmosphere. Please refer to the emission reduction calculation sheet attached with the report for the current monitoring period.                                                                          |

## 2 SAFEGUARDS

### 2.1 No Net Harm

The activities during construction, operation, and decommissioning of the project have been analysed to identify corresponding environmental and social impacts and assess their significance. Mitigation measures have been identified in the Environmental and Social Management Plan (ESMP) for negative impacts where institutional responsibilities for implementation and follow ups have been presented. The socioeconomic benefits of the project therefore outweigh the potential negative environmental and social impacts.

### 2.2 Local Stakeholder Consultation

A Local Stakeholder Consultation process was conducted in accordance with the VERRA-Verified Carbon Standard (VCS) requirements, towards the successful registration of emission reduction projects/programs aiming to generate carbon credits for the project activity. The details of the LSC were as follows:

Venue: CICB (Centre International de Conférence de Blitta)

Date: Tuesday, 5-July-2022

Time: 8:00 AM

The objective of the Stakeholders' Consultation event was to take due diligence of all comments, questions, and opinions in an open and transparent manner towards the project's economic, social, and environmental impacts, and sustainability. It was also to raise the stakeholders' awareness level about the procedure to register such projects under the VCS.

The grievance mechanism for the project complies with the following principles:

- Clarify at the outset what is the purpose of the procedure;
- Assure people that there will be neither cost nor retribution associated with lodging a grievance; and
- The entire process (i.e. how a complaint is received and reviewed, how decisions are made and what possibilities may exist for appeal) will be made as transparent as possible by putting it into written form, publicizing it and explaining it to the stakeholders.

To properly manage the potential impacts identified in the ESIA, a grievance mechanism based on the appointment of a Community Liaison officer in charge of managing community complaints has been designed.

The methods to access the grievance mechanism have been disclosed to the local communities. The grievance mechanism is publicized at the site gate, at the Local Community Chief's Offices, and at the Project Company's website<sup>3</sup>. All information about grievance procedures and grievance registers kept on site will be in English, but grievances forms can be filled in French.

However, since the project-affected communities have no or minimal access to the internet, but the great majority of them have a mobile phone, it's expected that a number of grievances will be received by phone. For such a reason the grievance mechanisms provided a phone number that is made available to the communities. Access to the mechanism is free of cost. No costs nor retribution is associated with lodging a grievance.

In addition to the explanation provided verbally, printed materials (e.g., posters) are affixed in prominent areas, providing the key facts about the mechanism and contact information, consisting of a dedicated phone number. Communication and awareness raising activities regarding the Grievance Mechanism are repeated at regular intervals.

The grievance log register for this monitoring period recorded a complaint raised by the workers of the O&M firm which has been taken into account by the O&M firm (Jakson Group). The firm was notified by the developer and has aligned their social management system with the National Labor Law, IFC PS2 & the Legal Social National System to ensure that the workers are provided with the necessary considerations in regard to all social aspects.

The grievance log register that shows how the ongoing communication has been recorded and addressed is given in Appendix II. For this monitoring period, there are no updates to the project design.

## 2.3 AFOLU-Specific Safeguards

Not applicable – this is a non-AFOLU project.

# 3 IMPLEMENTATION STATUS

## 3.1 Implementation Status of the Project Activity

The project activity consists of setting up a 70 MW solar PV plant in Blitta-Lasso in Togo, with 50 MW power already installed and a planned expansion of 20 MW. The commissioning date of

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<sup>3</sup> [https://ameapower.com/2022/wp-content/uploads/2022/10/15-16-Worker-and-community-grievance\\_vs\\_-\\_Copy.pdf](https://ameapower.com/2022/wp-content/uploads/2022/10/15-16-Worker-and-community-grievance_vs_-_Copy.pdf)

Phase 1 was on 12-June-2021 while the commissioning date of Phase 2 is planned for 01-December-2023. The technical details are as follows:

|                                  |                                                   |
|----------------------------------|---------------------------------------------------|
| <b>Solar Modules:</b>            |                                                   |
| Module Type                      | JKM390M-72-V                                      |
| Maximum Power (Pmax)             | 390/395/400 Wp                                    |
| Module technology                | Poly Crystalline                                  |
| No. of modules                   | 127,482                                           |
| <b>Inverters:</b>                |                                                   |
| Model                            | Sungrow SG250HX- 250KTL – H1 String or equivalent |
| No. of Inverters                 | 200                                               |
| Technical Lifetime of the plant: | 25 years                                          |

The project activity is in continuous operation since commissioning and there is no such incident reported during the current monitoring period which may impact the operation & capacity design of the project activity. Therefore, no events have happened that may impact the GHG emission reductions or removals and monitoring in this monitoring period.

## 3.2 Deviations

### 3.2.1 Methodology Deviations

Not applicable – there were no deviations applied during this monitoring period.

### 3.2.2 Project Description Deviations

Not applicable – there were no deviations applied during this monitoring period.

## 3.3 Grouped Projects

Not applicable – this project is not a grouped project.

# 4 DATA AND PARAMETERS

## 4.1 Data and Parameters Available at Validation

|                  |                           |
|------------------|---------------------------|
| Data / Parameter | EF <sub>grid, CM, y</sub> |
| Data unit        | tCO <sub>2</sub> /MWh     |

|                                                                                              |                                                                                                                   |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Description                                                                                  | Combined Margin CO <sub>2</sub> grid emission factor in the year y                                                |
| Source of data                                                                               | Standardized Baseline ASB0034-2021 Version 01.0 <sup>4</sup>                                                      |
| Value applied                                                                                | 0.573 tCO <sub>2</sub> /MWh                                                                                       |
| Justification of choice of data or description of measurement methods and procedures applied | The availability of a Standardized Baseline for the Grid Emission Factor for West African Power Pool Version 01.0 |
| Purpose of Data                                                                              | Calculation of baseline emissions                                                                                 |
| Comments                                                                                     | This parameter is fixed ex-ante for the entire crediting period                                                   |

## 4.2 Data and Parameters Monitored

| Data / Parameter                                                | EG <sub>PJ,y</sub>                                                                                              |                |                  |                   |               |                |      |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------|------------------|-------------------|---------------|----------------|------|
| Data unit                                                       | MWh/yr                                                                                                          |                |                  |                   |               |                |      |
| Description                                                     | Quantity of net electricity generation supplied to the grid during the monitoring period by the solar plant     |                |                  |                   |               |                |      |
| Source of data                                                  | Meter readings                                                                                                  |                |                  |                   |               |                |      |
| Description of measurement methods and procedures to be applied | Net electricity generation is electronically metered and stored under the Plant Manager's authority             |                |                  |                   |               |                |      |
| Frequency of monitoring/recording                               | Continuous measurement and monthly recordings                                                                   |                |                  |                   |               |                |      |
| Value monitored                                                 | Year                                                                                                            | Actual Value   | Corrected Value* |                   |               |                |      |
|                                                                 | 2021 (12-June-2021 to 31-December 2021)                                                                         | 42,607.20      | 42,586.47        |                   |               |                |      |
|                                                                 | 2022 (1-January-2022 to 31-December 2022)                                                                       | 81,269.20      | 81,269.20        |                   |               |                |      |
|                                                                 | Total                                                                                                           | 123,876.40     | 123,855.67       |                   |               |                |      |
|                                                                 | *Please refer to section '5.1 Baseline Emissions' for detailed explanation on Corrected Values.                 |                |                  |                   |               |                |      |
| Monitoring equipment                                            | <table> <tr> <th>Meter Description</th><th>Serial Number</th><th>Accuracy Class</th><th>Make</th></tr> </table> |                |                  | Meter Description | Serial Number | Accuracy Class | Make |
| Meter Description                                               | Serial Number                                                                                                   | Accuracy Class | Make             |                   |               |                |      |

<sup>4</sup> [https://cdm.unfccc.int/methodologies/standard\\_base/2015/sb166.html](https://cdm.unfccc.int/methodologies/standard_base/2015/sb166.html)



|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                           |          |      |            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------|------|------------|
|                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Power Trafo.1<br>Main     | 56198213 | 0.2S | Landis+Gyr |
|                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Power Trafo.1<br>Check    | 56198214 | 0.2S | Landis+Gyr |
|                                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Power Trafo.2<br>Main     | 56198215 | 0.2S | Landis+Gyr |
|                                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Power Trafo.2<br>Check    | 56198216 | 0.2S | Landis+Gyr |
|                                | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Blitta feeder<br>Main     | 56198203 | 0.2S | Landis+Gyr |
|                                | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Blitta feeder<br>Check    | 56198217 | 0.2S | Landis+Gyr |
|                                | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Auxiliary Trafo.<br>Main  | 56198221 | 0.2S | Landis+Gyr |
|                                | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Auxiliary Trafo.<br>Check | 56198222 | 0.2S | Landis+Gyr |
| QA/QC procedures to be applied | <p>The calibration of the meters is carried out on an annual basis as agreed by all parties and as indicated in the PPA.</p> <p>The calibration is carried out by a 3<sup>rd</sup> party every year, and copies of the calibration certificates are signed by all parties.</p> <p>For this monitoring period (June-2021 to December-2022), three calibration certificates are provided that cover this period. Copies of those certificates and the calibration dates are provided in Appendix IV.</p> |                           |          |      |            |
| Purpose of the data            | Calculation of baseline emissions                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |          |      |            |
| Calculation method             | <i>Not applicable</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                           |          |      |            |
| Comments                       | <p>Data will be archived in paper &amp; electronically for a period of 2 years beyond the end of crediting period or of the last issuance of credits for this project activity, whichever occurs later.</p>                                                                                                                                                                                                                                                                                            |                           |          |      |            |

### 4.3 Monitoring Plan

The proposed Monitoring Plan describes the Management System and Procedures to be developed and implemented by the project proponent for this project in order to ensure consistency with the methodology ACM0002: Grid-connected electricity generation from renewable sources — Version 20.0, which is applied in the context of the project activity.

The developed monitoring plan contains the following sections:

1. Monitoring parameters
2. Monitoring Points
3. Monitoring equipment
4. Management structure
5. Monitoring guidelines, procedures and documentation
6. Deviation Control Mechanism
7. Regular staff training

All data required for the calculation of emission reductions associated with the project activity is monitored in accordance with the approved methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0 and will be archived electronically and on paper throughout the crediting period and up to 2 years thereafter.

A Monitoring Management and Quality Assurance System (MMQAS) is developed and implemented to ensure that the results from the project activity are reliable and transparent.

The complete MMQAS, including data and documents from project operation, is open for review to the VVBs contracted for verification.

### **Monitoring Parameters**

The tables under section 5.1 provide information on the required monitoring parameters as per the approved methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0 and referred tools, required for the ex-post calculation of project emissions associated with the project activity.

### **Monitoring Points**

Please refer to the Single line diagram in Appendix III.

### **Monitoring equipment**

The physical monitoring system consists of the following meter equipment:

Electricity meter: AMEA Solar Togo SAU to monitor the quantity of net electricity generation supplied to the grid in year  $y$  by the solar plant ( $EG_{PJ,y}$ )

As shown in the calculations of section 4.4 and according to the methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 20.0, baseline emissions are calculated by multiplying the quantity of net electricity generation supplied to the grid in year  $y$  by the solar plant ( $EG_{PJ,y}$  with) with the grid emission factor in the year  $y$  ( $EF_{grid, CM, y}$ )

### **Management structure**

A VCS team is designated to be responsible for carrying out the required monitoring tasks for the project activity. Both management people and technical workers are present in the VCS team. All of the team members have passed internal monitoring training.

The management structure consists of 3 levels to guarantee a step-by-step management of the project activity, which casts concrete responsibilities on the appointed person who will be actually in charge of a specific task.

The 3 levels refer to:

1. Solar Plant Manager
2. VCS project manager
3. VCS staff

Before the start of the crediting period, clear roles and responsibilities are assigned to all staff involved in the project activity. A VCS Project Manager is appointed who is responsible for monitoring activities and data management associated with the project activity. All operators involved in the collection of data and records are coordinated by the VCS Manager.

### **Monitoring guidelines, procedures and documentation**

The introduced Monitoring Management and Quality Assurance System is built on written guidelines and procedures provided to the operational VCS team staff and to the local VCS management.

The procedures were developed by the external consultant according to the approved methodology ACM0002: Grid-connected electricity generation from renewable sources — Version 20.0. All tasks under the monitoring plan are carried out strictly according to the provided guidelines and procedures.

The Monitoring Management and Quality Assurance System is designed to enable VCS staff members to perform the required tasks following the given guidelines and procedures. Providing documentation designed specifically for the required tasks will strongly benefit availability and transparency of data streams and processes which are relevant for the VCS project.

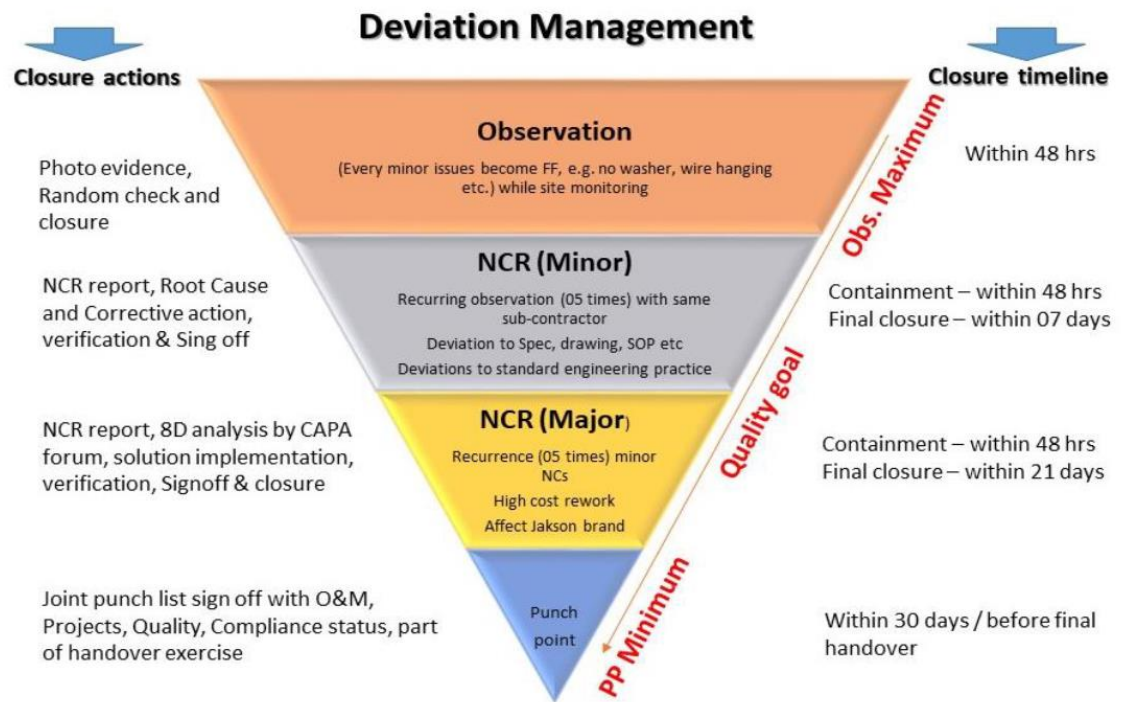
The VCS Project Manager carries out frequent performance and data cross-checking, to ensure that staff are following the Monitoring Management and Quality Assurance System.

All documents filled out and data acquired during the project will be archived separately for backup.

### **Deviation Control Mechanism**

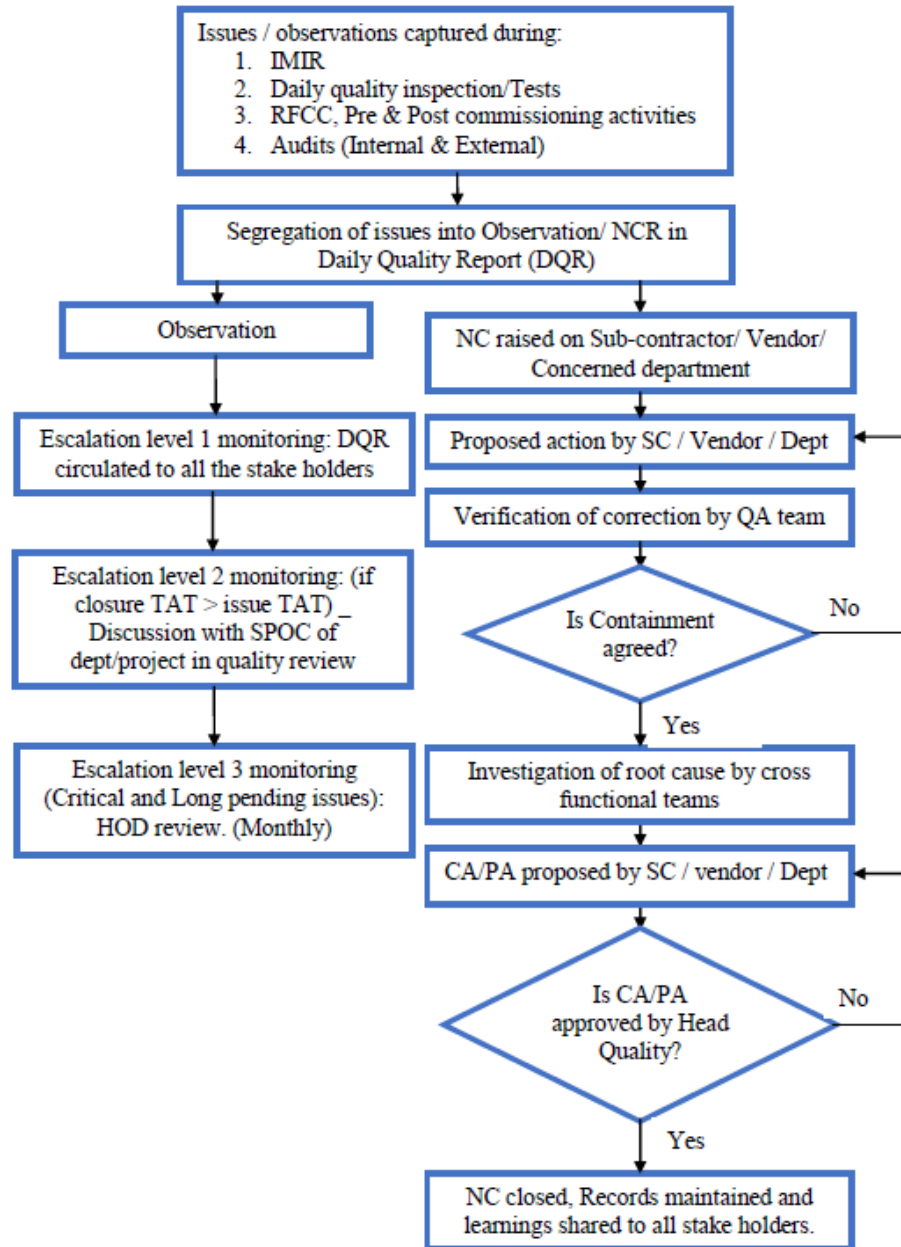
Daily inspection is carried out by the quality team to verify that the activity is performed as per the First Off/SOP (Standard Operating Procedure) or as per the standard. Any deviation observed during inspection is recorded in the DQR (Daily Quality Report) and circulated to all

stakeholders on a daily basis. Deviations are classified into observations, NC (Non-conformance) (Minor & Major) and Punch points as shown below:



If closure TAT (Turn Around Time) is more than the defined timeline, points are discussed in quality review meetings. Process flow of deviation control mechanism and NC management is shown in the below flowchart:

#### Deviation Control Mechanism:



The list of abbreviations is provided below:

| Abbreviation | Definition                          |
|--------------|-------------------------------------|
| IMIR         | Inward Material Inspection Report   |
| RFCC         | Request For Commissioning Clearance |
| SPOC         | Single Point of Contact             |
| CA/PA        | Corrective Action/Preventive Action |
| SC           | Sub Contractor                      |
| HOD          | Head of Department                  |

|     |                        |
|-----|------------------------|
| NCR | Non-Conformance Report |
|-----|------------------------|

**Regular Staff Training**

The Monitoring Management and Quality Assurance System is currently being implemented and maintained. In order to qualify local operational staff to carry out required regular tasks professional training is provided, both in class and practical application, to the designated VCS team staff. The regular staff training is provided at least once a year for the duration of the project.

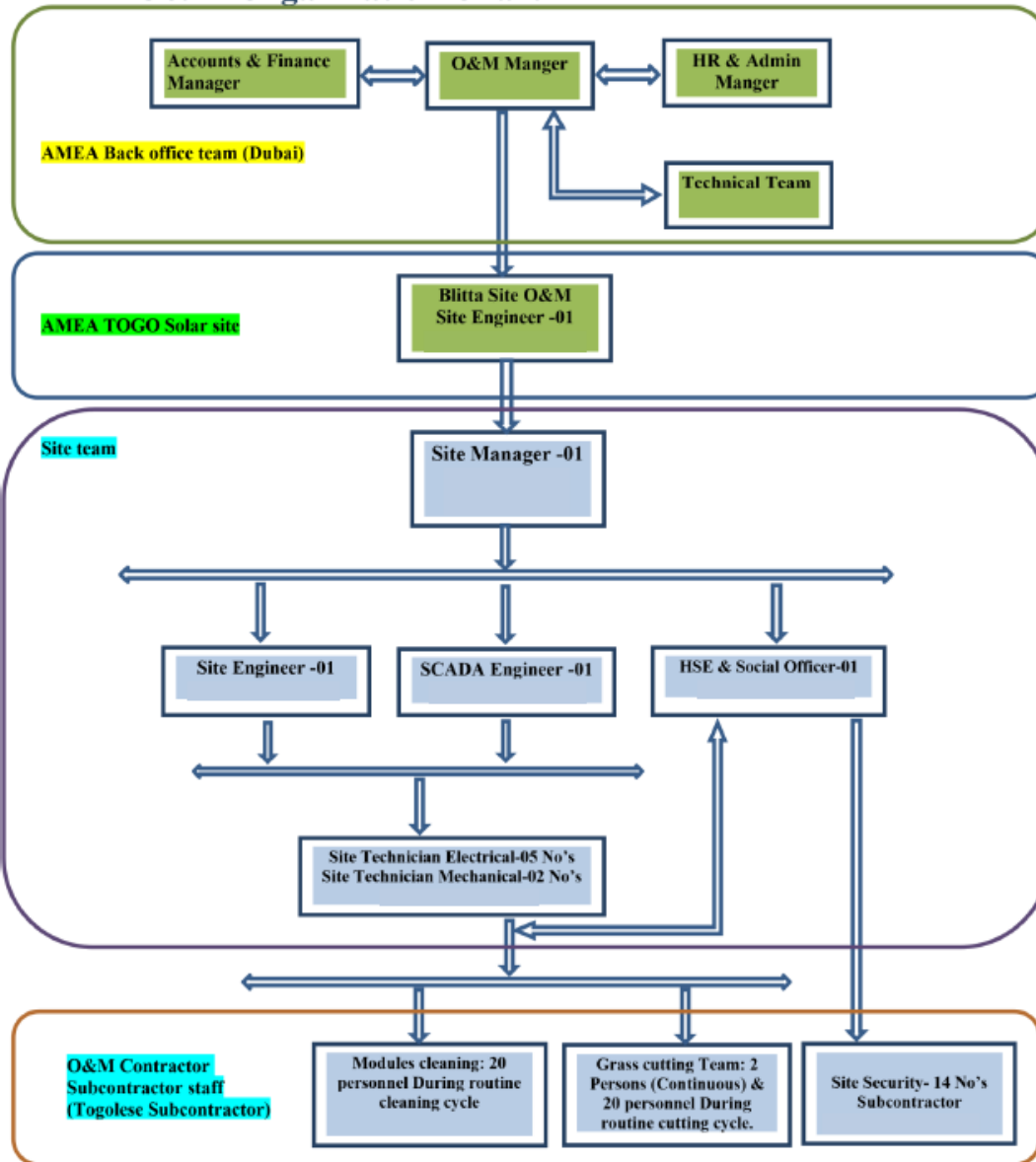
Only trained staff will be members of the VCS team, and only trained VCS staff members will be allowed to perform any tasks related to monitoring activities.

The VCS Manager is permitted to train new staff. By this it will be ensured that operational VCS staff are qualified to carry out the relevant tasks under the Monitoring Management and Quality Assurance System.

This monitoring report has been prepared based on the results of the current monitoring period. An overall QA/QC on the Monitoring Spreadsheet and Monitoring Report is carried out by an external Consultant before the documents are submitted to the selected VVB.

The O&M Organization Chart is shown below:

**TOGO, BLITTA**

**O&M Organization Chart**


## 5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

### 5.1 Baseline Emissions

As per the procedure established in the registered PD,

The baseline emissions are to be calculated as follows:

$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$  Where:

$BE_y$  = Baseline emissions in year y (tCO<sub>2</sub>/yr)

$EG_{PJ,y}$  = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

For this monitoring period, the DGR values are considered for the calculations of the emission reductions as they are the more conservative values.

As the first calibration of the meters took place on 12-October-2020 and the second one on 22-November-2021, there is a delay of 42 days (including both dates)

According to the 'CDM validation and verification standard for project activities', 366(a), if the meters' calibration is delayed, then the manufacturer's maximum permissible error must be applied to the metered values for that period, provided the error observed during calibration is below the maximum permissible error.

The calibration certificates provided in the MR show that the error observed during calibration was below the maximum permissible error which is  $\pm 0.2\%$  for 0.2S class meters (as shown in calibration certificate). This error being applied to the delayed 42 days and the difference is then subtracted from the total actual generated value.

$EF_{grid,CM,y}$  = Combined margin CO<sub>2</sub> emission factor for grid connected power generation in year y calculated using the latest version of "TOOL07: Tool to calculate the emission factor for an electricity system" (tCO<sub>2</sub>/MWh) referring to the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool, Version 01.0.

#### Calculation of $EG_{PJ,y}$

In case of Greenfield power plant development, the quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y, according to the methodology is calculated as follows:



$EG_{PJ,y} = EG_{facility,y}$  Where:

$EG_{PJ,y}$  = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$  = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

#### Calculation of $EF_{grid,CM,y}$

As stated before, the “TOOL07: Tool to calculate the emission factor for an electricity system” allows referring to the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool, Version 01.0.

The ASB0034-2021 Standardized Baseline states that for wind and solar power generation project activities, the  $EF_{grid,CM,y}$  to be used is 0.573 tCO<sub>2</sub>/MWh.

The total quantity of net electricity supplied to the grid for the current monitoring period:

$$EG_{PJ,y} = 123,876.40 \text{ MWh}$$

The corrected net electricity supplied to the grid for the current monitoring period (as explained earlier in the section):

$$EG_{PJ,y} = 123,855.67 \text{ MWh}$$

The Grid Emission Factor is:

$$EF_{grid,CM,y} = 0.573 \text{ tCO}_2/\text{MWh}$$

The baseline emissions of the project activity are:

$$BE_y = 123,855.67 \text{ MWh} * 0.573 \text{ tCO}_2/\text{MWh} = 70,969 \text{ tCO}_2\text{e (Round down value)}$$

## 5.2 Project Emissions

The project activity involves renewable power generation. Therefore,  $PE_y = 0$

## 5.3 Leakage

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g., extraction, processing, transport etc.) are not calculated.

## 5.4 Net GHG Emission Reductions and Removals

As per the applied methodology, emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where,

$ER_y$  = Emission Reduction in tCO<sub>2</sub>e/year

$BE_y$  = Baseline emission in tCO<sub>2</sub>e/year

$PE_y$  = Project emissions in tCO<sub>2</sub>e/year

$ER_y = 70,969 \text{ tCO}_2\text{e} - 0 \text{ tCO}_2\text{e} = 70,969 \text{ tCO}_2\text{e}$

| Year                                              | Baseline emissions or removals (tCO <sub>2</sub> e) | Project emissions or removals (tCO <sub>2</sub> e) | Leakage emissions (tCO <sub>2</sub> e) | Net GHG emission reductions or removals (tCO <sub>2</sub> e) |
|---------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|----------------------------------------|--------------------------------------------------------------|
| Year 2021<br>(12-June-2021 to 31-December-2021)   | 24,402                                              | 0                                                  | 0                                      | 24,402                                                       |
| Year 2022<br>(1-January-2022 to 31-December-2022) | 46,567                                              | 0                                                  | 0                                      | 46,567                                                       |
| <b>Total</b>                                      | <b>70,969</b>                                       | <b>0</b>                                           | <b>0</b>                               | <b>70,969</b>                                                |

| <u>Ex-ante emissions reductions /removals</u> | <u>Achieved emissions reductions /removals</u> | <u>Percent difference</u> | <u>Justification for the difference</u>                                                                                                              |
|-----------------------------------------------|------------------------------------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 110,807                                       | 70,969                                         | -35.95%                   | The ex-ante emissions were calculated based on 70 MW. The current monitoring period accounts for 50 MW as the 20 MW is currently under construction. |

# APPENDIX I: SDG CONTRIBUTIONS

AMEA Solar Togo SAU built the first clinic in Blitta, and this has improved the standard of living within the village. The number of patients visiting the newly constructed clinic increased as there was no clinic previously. This contributes towards SDG 3: Good Health and Well-being that aims to focus on key targets that boost the health of a country's overall population. The following shows the number of patients visiting the clinic for consultations.



Visitors of AMEA Maternity and Peditry

| Medical Consultation              |                               |                  |       |       |
|-----------------------------------|-------------------------------|------------------|-------|-------|
| MONTH                             | PERIOD                        | NUMBER OF PEOPLE |       |       |
| OCTOBER                           | From 06.10.2022 to 31.10.2022 | 490              |       |       |
| NOVEMBER                          | From 01.11.2022 to 30.11.2022 | 582              |       |       |
| DECEMBER                          | From 01.12.2022 to 09.12.2022 | 249              |       |       |
| Maternity (Prenatal consultation) |                               |                  |       |       |
| OCTOBER                           | From 06.10.2022 to 31.10.2022 | 100              |       |       |
| NOVEMBER                          | From 01.11.2022 to 30.11.2022 | 106              |       |       |
| DECEMBER                          | From 01.12.2022 to 09.12.2022 | 49               |       |       |
| Birth                             |                               |                  |       |       |
|                                   |                               | Boys             | Girls | Total |
| OCTOBER                           | From 06.10.2022 to 31.10.2022 | 6                | 12    | 18    |
| NOVEMBER                          | From 01.11.2022 to 30.11.2022 | 5                | 12    | 17    |
| DECEMBER                          | From 01.12.2022 to 09.12.2022 | 4                | 6     | 10    |

Blitta village le 06/02/2023



AMEA Solar Togo SAU has built three (3) primary schools and renovated the four (4) existing primary schools with the necessary equipment and furniture. The number of students going to primary schools annually has increased. These activities contribute towards SDG 4: Quality Education that ensures inclusive and equitable quality education and promotes lifelong learning opportunities for all. The following shows the number of students admitted to each of the schools.

### EPP BLITTA CENTRAL D (TOMEGBE)

| Year    | 2022-2023 |       |       |      |       |       |      |       |       |      |       |       |
|---------|-----------|-------|-------|------|-------|-------|------|-------|-------|------|-------|-------|
| CLASSES | CE1       |       |       | CE2  |       |       | CM1  |       |       | CM2  |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS | GIRLS | TOTAL | BOYS | GIRLS | TOTAL | BOYS | GIRLS | TOTAL |
| NUMBER  | 19        | 25    | 44    | 16   | 27    | 43    | 22   | 19    | 41    | 10   | 20    | 30    |

| Year    | 2021-2022 |       |       |      |       |       |      |       |       |      |       |       |
|---------|-----------|-------|-------|------|-------|-------|------|-------|-------|------|-------|-------|
| CLASSES | CE1       |       |       | CE2  |       |       | CM1  |       |       | CM2  |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS | GIRLS | TOTAL | BOYS | GIRLS | TOTAL | BOYS | GIRLS | TOTAL |
| NUMBER  | 16        | 17    | 33    | 17   | 19    | 36    | 18   | 18    | 36    | 10   | 16    | 26    |

| Year    | 2020-2021 |       |       |      |       |       |      |       |       |      |       |       |
|---------|-----------|-------|-------|------|-------|-------|------|-------|-------|------|-------|-------|
| CLASSES | CE1       |       |       | CE2  |       |       | CM1  |       |       | CM2  |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS | GIRLS | TOTAL | BOYS | GIRLS | TOTAL | BOYS | GIRLS | TOTAL |
| NUMBER  | 15        | 24    | 39    | 19   | 20    | 39    | 16   | 21    | 37    | 16   | 14    | 30    |

Fait à Blitta le 06/02/23



Modèle NABIDA

### LYCEE BLITTA CARREFOUR

| Year    | 2022-2023 |       |       |        |       |       |        |       |       |        |       |       |
|---------|-----------|-------|-------|--------|-------|-------|--------|-------|-------|--------|-------|-------|
| CLASSES | Siem A    |       |       | Siem B |       |       | 3iem A |       |       | 3iem B |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL |
| NUMBER  | 46        | 58    | 104   | 47     | 57    | 104   | 43     | 43    | 86    | 45     | 40    | 85    |

| Year    | 2021-2022 |       |       |        |       |       |        |       |       |        |       |       |
|---------|-----------|-------|-------|--------|-------|-------|--------|-------|-------|--------|-------|-------|
| CLASSES | Siem A    |       |       | Siem B |       |       | 3iem A |       |       | 3iem B |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL |
| NUMBER  | 44        | 14    | 58    | 44     | 14    | 58    | 44     | 20    | 64    | 44     | 20    | 64    |

Fait à Blitta, le 06 Février 2023

Le Proviseur,



KOUSSANTHA Bougoundjama

### EPP KATAKPE

| Year    | 2022-2023 |       |       |        |       |       |        |       |       |        |       |       |
|---------|-----------|-------|-------|--------|-------|-------|--------|-------|-------|--------|-------|-------|
| CLASSES | CP 1&2    |       |       | CE 1&2 |       |       | CM 1&2 |       |       | JARDIN |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL |
| NUMBER  | 17        | 9     | 26    | 11     | 12    | 23    | 14     | 8     | 22    | 11     | 12    | 23    |

| Year    | 2021-2022 |       |       |        |       |       |        |       |       |        |       |       |
|---------|-----------|-------|-------|--------|-------|-------|--------|-------|-------|--------|-------|-------|
| CLASSES | CP 1&2    |       |       | CE 1&2 |       |       | CM 1&2 |       |       | JARDIN |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL |
| NUMBER  | 13        | 13    | 26    | 11     | 11    | 22    | 14     | 5     | 19    | 11     | 9     | 20    |

Le 6 102/2023



TOTA Djimba

### EPP LAMAKAZA B

| Year    | 2022-2023 |       |       |      |       |       |      |       |       |
|---------|-----------|-------|-------|------|-------|-------|------|-------|-------|
| CLASSES | CP1       |       |       | CE2  |       |       | CM1  |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS | GIRLS | TOTAL | BOYS | GIRLS | TOTAL |
| NUMBER  | 40        | 40    | 80    | 33   | 31    | 64    | 29   | 30    | 59    |

| Year    | 2021-2022 |       |       |      |       |       |      |       |       |
|---------|-----------|-------|-------|------|-------|-------|------|-------|-------|
| CLASSES | CE1       |       |       | CE2  |       |       | CM1  |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS | GIRLS | TOTAL | BOYS | GIRLS | TOTAL |
| NUMBER  | 36        | 29    | 65    | 26   | 26    | 52    | 27   | 36    | 63    |

| Year    | 2020-2021 |       |       |      |       |       |      |       |       |
|---------|-----------|-------|-------|------|-------|-------|------|-------|-------|
| CLASSES | CE1       |       |       | CE2  |       |       | CM1  |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS | GIRLS | TOTAL | BOYS | GIRLS | TOTAL |
| NUMBER  | 21        | 21    | 42    | 26   | 28    | 54    | 17   | 23    | 40    |

06/02/2023  
LE DIRECTEUR  
BATCHASSI N'na

### EPP KOKONADE

| Year    | 2022-2023 |       |       |        |       |       |        |       |       |        |       |       |
|---------|-----------|-------|-------|--------|-------|-------|--------|-------|-------|--------|-------|-------|
| CLASSES | CP 1&2    |       |       | CE 1&2 |       |       | CM 1&2 |       |       | JARDIN |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL |
| NUMBER  | 14        | 13    | 27    | 16     | 13    | 29    | 9      | 7     | 16    | 9      | 13    | 22    |

| Year    | 2021-2022 |       |       |        |       |       |        |       |       |        |       |       |
|---------|-----------|-------|-------|--------|-------|-------|--------|-------|-------|--------|-------|-------|
| CLASSES | CP 1&2    |       |       | CE 1&2 |       |       | CM 1&2 |       |       | JARDIN |       |       |
| GENDER  | BOYS      | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL | BOYS   | GIRLS | TOTAL |
| NUMBER  | 25        | 9     | 34    | 9      | 17    | 26    | 11     | 7     | 18    | 15     | 9     | 24    |

Fait à Blitta le 06/02/23  
LE DIRECTEUR  
MINZA Mewélouwé

## APPENDIX II: STAKEHOLDER ENGAGEMENT

The grievances log and the stakeholder engagement log for the current monitoring period are shown below:

### Social – grievances log:

| # |                | GRIEVANCE         |                                |              |                                                                         | RECORD              |                     |                   |
|---|----------------|-------------------|--------------------------------|--------------|-------------------------------------------------------------------------|---------------------|---------------------|-------------------|
| N | Reference      | Date of complaint | Complainer name                | Access point | Brief description                                                       | Date of acknowledge | Receptor name       | Type of grievance |
| 1 | BLIT-GRIEV-001 | 17-November-2021  | JAKSON<br>Togolese O&M workers | Telephone    | The workers are claiming the regularization of their working conditions | 17-November-2021    | Abdou-Raouf Damtare | Worker            |

| ANALYSIS           |                         |                                                                                                                                                                                       |                         | IMPLEMENTATION                                                                                                                                                                                                                                                                               |               |
|--------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| ASSIGNED OWNERSHIP | SEVERITY CLASSIFICATION | BRIEF DESCRIPTION OF EVALUATION                                                                                                                                                       | TIMELINE FOR RESOLUTION | ACTIONS TAKEN                                                                                                                                                                                                                                                                                | CLOSURE DATE  |
| Jakson             | HIGH                    | The O&M firm was not taking the workers' social dilemmas and complaints into serious consideration. The firm must ensure to be in compliance with the National Labor Law and IFC PS2. | 30-November-2021        | The O&M firm (Jakson Group) was notified by the developer and has aligned their social management system with the National Labor Law, IFC PS2 & the Legal social National System to ensure that the workers are provided with the necessary considerations in regards to all social aspects. | 31-March-2022 |

| ASSESSMENT POST-CLOSURE            |                                    |                        |        |
|------------------------------------|------------------------------------|------------------------|--------|
| LEVEL OF SATISFACTION WITH PROCESS | LEVEL OF SATISFACTION WITH OUTCOME | COMMENTS               | STATUS |
| SATISFIED                          | SATISFIED                          | CNSS has been provided | CLOSED |

#### Stakeholder engagement log:

| RECORD |              |                |             | EVENT   |                       |                              |                                                                          |          |
|--------|--------------|----------------|-------------|---------|-----------------------|------------------------------|--------------------------------------------------------------------------|----------|
| #      | REFERENCE    | DATE           | LOGGED BY   | EVENT   | TOPICS DISCUSSED      | PARTIES STAKEHOLDER INVOLVED | MAIN DISCUSSION AND FOLLOW-UP ACTIONS                                    | COMMENTS |
| 1      | BLIT-SEL-001 | 4-August-2021  | Hervé Rioux | Visit   | Security of the site  | Togolese Army                | Togolese army arrived on site to see dispositions taken for security     |          |
| 2      | BLIT-SEL-002 | 5-August-2021  | Hervé Rioux | Control | Environmental control | A.N.G. E                     | The national environmental agency arrived to check environmental aspects |          |
| 3      | BLIT-SEL-003 | 7-August-2021  | Hervé Rioux | Visit   | Security of the site  | Togolese Army                | Togolese army arrived on site to see dispositions taken for security     |          |
| 4      | BLIT-SEL-004 | 17-August-2021 | Abdou-raouf | Visit   | Site works            | CEET                         | CEET workers arrived to check the how things are working on site         |          |

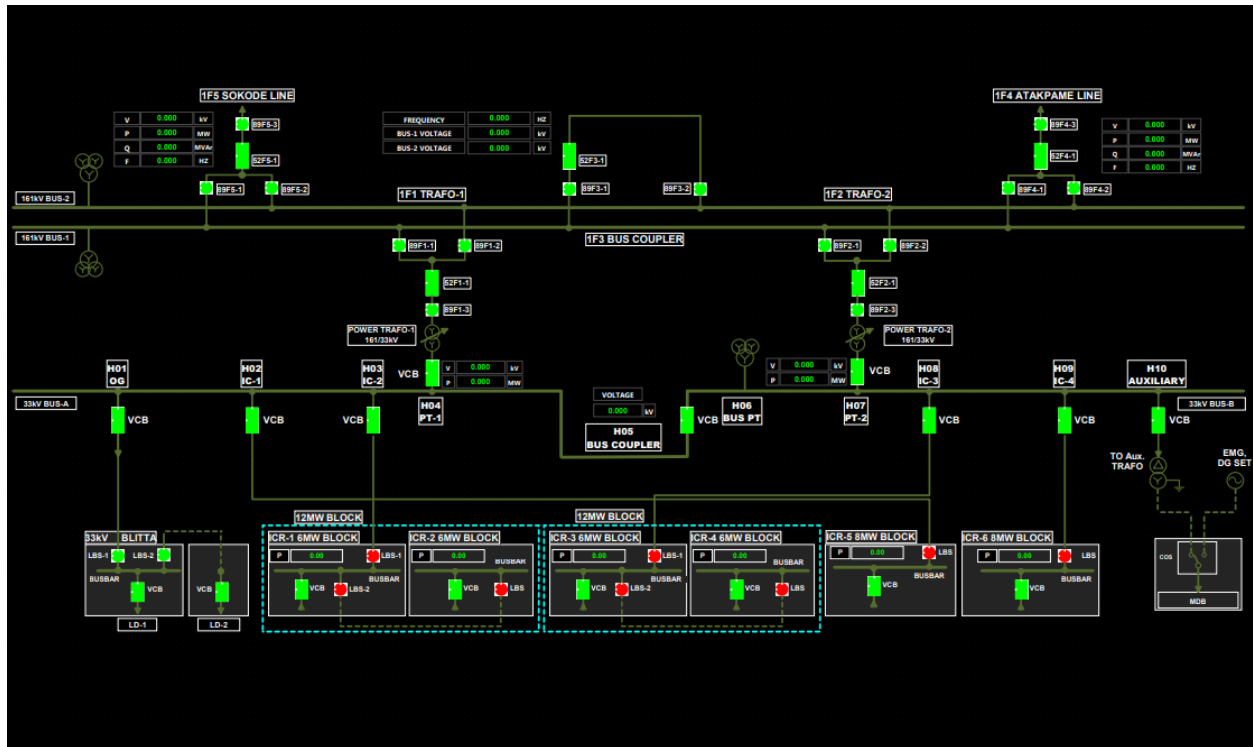


|    |              |                   |                      |       |                      |               |                                                                                                            |  |
|----|--------------|-------------------|----------------------|-------|----------------------|---------------|------------------------------------------------------------------------------------------------------------|--|
| 5  | BLIT-SEL-005 | 18-August-2021    | Abdou-raouf          | Visit | Security of the site | Togolese Army | Togolese army arrived on site to see dispositions taken for security                                       |  |
| 6  | BLIT-SEL-006 | 26-August-2021    | Abdou-raouf          | Visit | Site works           | KYA Energy    | KYA Energy workers arrived to check the how things are working on site                                     |  |
| 7  | BLIT-SEL-007 | 27-August-2021    | Abdou-raouf          | Visit | Security of the site | Togolese Army | Togolese army arrived on site to see dispositions taken for security                                       |  |
| 8  | BLIT-SEL-008 | 1-September-2021  | Abdou-raouf          | Visit | Security of the site | Togolese Army | Togolese army arrived on site to see dispositions taken for security                                       |  |
| 9  | BLIT-SEL-009 | 10-September-2021 | Abdou-raouf          | Visit | Site works           | CEET          | CEET workers arrived to check the how things are working on site                                           |  |
| 10 | BLIT-SEL-010 | 16-September-2021 | Abdou-raouf          | Visit | Pending points       | ARSE          | ARSE workers came to see what have been done regarding all the points they highlighted during construction |  |
| 11 | BLIT-SEL-011 | 22-September-2021 | KARUNA & Abdou-raouf | Visit | Security of the site | Togolese Army | Togolese army arrived on site to see dispositions taken for security                                       |  |
| 12 | BLIT-SEL-011 | 22-September-2021 | KARUNA & Abdou-raouf | Visit | Security of the site | Togolese Army | Togolese army arrived on site to see dispositions taken for security                                       |  |

|    |              |                  |               |       |                                      |                          |                                                                                                                                           |                                             |
|----|--------------|------------------|---------------|-------|--------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 13 | BLIT-SEL-013 | 27-October-2021  | Abdou-raouf   | Visit | Pending points                       | ARSE                     | ARSE workers came to see what have been done regarding all the points they highlighted during construction                                |                                             |
| 14 | BLIT-SEL-014 | 23-November-2021 | Hervé & Raouf | Visit | Phase 3 financial topics             | BOAD                     | BOAD came on site to see what is done and see how they can finance the phase 3                                                            |                                             |
| 15 | BLIT-SEL-015 | 09-December-2021 | Hervé & Raouf | Visit | Plant working                        | BOAD                     | Technical team came to see learn how the plant is working                                                                                 |                                             |
| 16 | BLIT-SEL-016 | 10-December-2021 | Hervé & Raouf | Visit | Plant working                        | BOAD                     | Technical team came to see learn how the plant is working                                                                                 |                                             |
| 17 | BLIT-SEL-017 | 10-December-2021 | Hervé & Raouf | Visit | Plant working                        | BOAD                     | Technical team came to see learn how the plant is working                                                                                 |                                             |
| 18 | BLIT-SEL-018 | 17-February-2022 | Hervé & Raouf | Visit | Pending points and operation license | ARSE                     | Technical and legal team of ARSE came on site to ensure that AMEA comply with the pending points in order to obtain the operating license | Closing of those points by AMEA are ongoing |
| 19 | BLIT-SEL-019 | 30-March-2022    | Hervé & Raouf | Visit | Finance audit                        | Deloitte & ACGroup-afric | The financial audit team arrived on site with the Finance Manager of AMEA to visit the site and also discuss finance purpose              |                                             |

|    |              |               |               |                     |                                              |                   |                                                                                                                                          |                                   |
|----|--------------|---------------|---------------|---------------------|----------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 20 | BLIT-SEL-020 | 30-March-2022 | Hervé & Raouf | Visit               | Environmental control                        | ANGE              | The national environmental agency arrived to check if AMEA is respecting all the environmental aspects                                   |                                   |
| 21 | BLIT-SEL-021 | 14-April-2022 | Hervé & Raouf | Visit               | Pending points and operation license         | ARSE              | Technical and legal team of ARSE came on site to control the pending points in order to obtain the operating license                     | All the points are closed by AMEA |
| 22 | BLIT-SEL-022 | 12-May-2022   | Hervé & Raouf | Work on Fiber Optic | Fiber Optic checking                         | CEB               | CEB team came on site to check and repair fiber optic cable                                                                              |                                   |
| 23 | BLIT-SEL-023 | 24-May-2022   | Raouf         | Work on Fiber Optic | Fiber Optic checking                         | CEB               | CEB team came on site to check and repair fiber optic cable                                                                              |                                   |
| 24 | BLIT-SEL-024 | 15-June-2022  | Hervé & Raouf | Meeting             | Annual review PR of the plant                | ARSE, CEET, AT2ER | The national agencies arrived on site to make review of one year production and PR checking by referring to the power purchase agreement |                                   |
| 25 | BLIT-SEL-025 | 17-June-2022  | Hervé & Raouf | Meeting             | Validation of the ESIA of the phase 3        | ANGE              | The national environmental agency made a meeting with all the stakeholders for the validation of the ESIA of phase 3                     |                                   |
| 26 | BLIT-SEL-026 | 7-July-2022   | Raouf         | Control             | Air quality, water quality and noise quality | ANGE              | The national environmental agency arrived to check environmental aspects                                                                 | Report received and all good      |

|    |              |                  |               |       |                                       |                           |                                                                                                                                                                                                           |                                  |
|----|--------------|------------------|---------------|-------|---------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 27 | BLIT-SEL-027 | 18-August-2022   | Hervé & Raouf | Visit | Working conditions of the plant       | CEB, CEET, DGE            | We discussed about the working conditions of the site and the different issues face with CEB regarding the voltage and the frequency                                                                      |                                  |
| 28 | BLIT-SEL-028 | 31-October-2022  | Hervé & Raouf | Visit | Working conditions of the plant       | CEET                      | The DG of CEET arrived to visit the site and CEET workers also. We discussed also about the working conditions of the site and the different issues face with CEB regarding the voltage and the frequency |                                  |
| 29 | BLIT-SEL-029 | 13-December-2022 | Hervé & Raouf | Visit | Working Conditions of workers on site | Togolese labor inspectors | The labor inspectors arrived to see if AMEA is respecting the Togolese labor law on site                                                                                                                  | No major concerns were reported. |



| Serial Number | Calibration Dates |
|---------------|-------------------|
| 56198213      | 12-October-2020   |
|               | 22-November-2021  |
|               | 25-October-2022   |
| 56198214      | 12-October-2020   |
|               | 22-November-2021  |
|               | 25-October-2022   |
| 56198215      | 12-October-2020   |
|               | 22-November-2021  |
|               | 25-October-2022   |

|          |                  |
|----------|------------------|
| 56198216 | 12-October-2020  |
|          | 22-November-2021 |
|          | 25-October-2022  |
| 56198203 | 12-October-2020  |
|          | 22-November-2021 |
|          | 25-October-2022  |
| 56198217 | 12-October-2020  |
|          | 22-November-2021 |
|          | 25-October-2022  |
| 56198221 | 12-October-2020  |
|          | 22-November-2021 |
|          | 25-October-2022  |
| 56198222 | 12-October-2020  |
|          | 22-November-2021 |
|          | 25-October-2022  |